PalmSource™ 99

New Tools for Building Conduits

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Agenda

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- PDDirect Overview
  - COM/DCOM Conduits
- System Architecture
- How to use PDDirect

- Demo
  - Look at some code
  - Build a conduit

COM/DCOM Conduits

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- PDDirect is a software toolkit
  - Authoring of conduits using any COM enabled programming language
  - PDDirect provides you with a simple and easy way to directly access the Palm from almost any Windows application
- Implemented using Microsoft COM
- Extends the Sync Manager API
  - Categories object, Iterator object, Multiple DBs
  - Many unique features
- Powerful, but easy to learn and use
  - Minimal lines of code, few objects

Programming Languages



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Any COM Compliant Language

- Visual Basic (VB)
- Visual Basic for Applications (VBA)
- Delphi
- Borland C/C++
- Java
- Microsoft Visual C/C++, versions 4.x-6.x
  - (MFC is not required!)

PDDirect Unique Features

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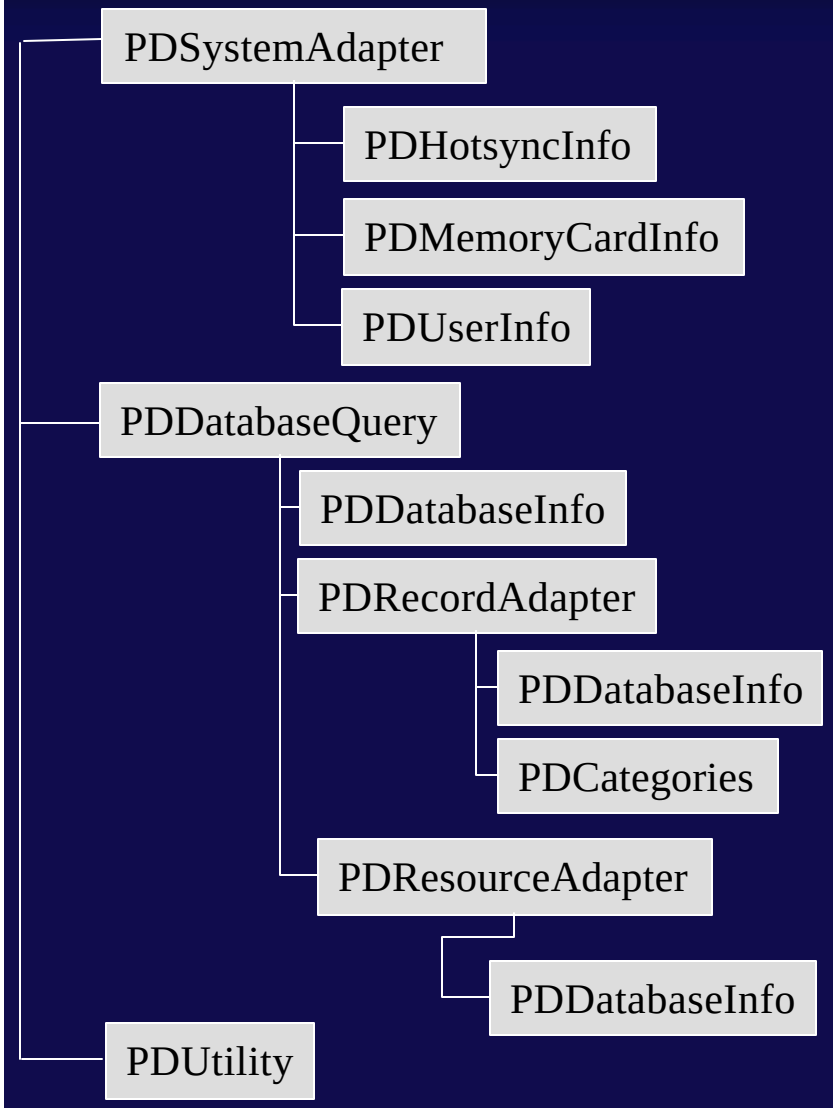
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- Debug your conduits in real-time
- Lightweight controls written in C++/ATL without MFC
- HotSync® keep-alive (SyncYieldCycles) calls are not necessary
- Multi-threading support
- Multi-database Support
- Multiple abstraction models supported
- DCOM support
- Support for HotSync 2.x and higher
- Full Sync Manager 3.0 support
- Partial Conduit Manager support
- Support any COM compliant programming language
- PDDirect permits you to build conduits in multiple forms

Object Hierarchy

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PDSystemAdapter: Access to system methods and information

PDHotsyncInfo: Information supplied by the HotSync to the conduit

PDMemoryCardInfo: Memory card information

PDUserInfo: User information

PDDatabaseQuery: Database management, access and information

PDDatabaseinfo: Individual database information

PDRecordAdapter: Methods and properties for access to a single open database

PDCategories: Wrapper object for the AppInfo block permitting easy access to the database categories

AppInfo: block permitting easy access to the database categories

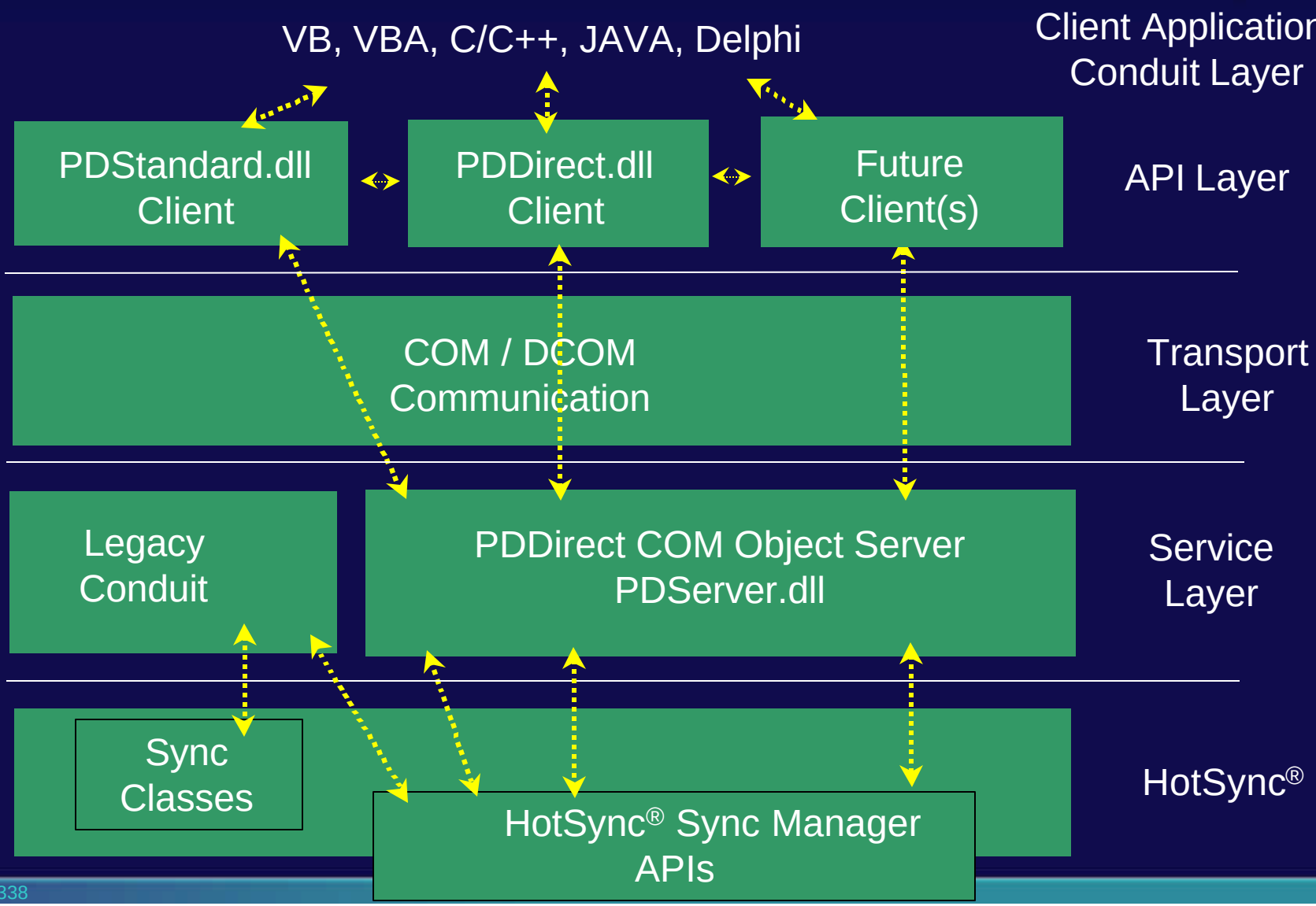
PDResourceAdapter: Methods and properties for access to a single open resource database

PDUtility: Byte array access, print formatting and other useful utility methods

PDDirect System Architecture

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PDDirect Functional Overview



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- The server DLL "wraps" the Sync Manager API with a COM interface, while the client DLLs expose that COM interface to you through an easy to understand object model
- Server:
  - PDServer.dll is an object server that will remain active as long as there are active objects or as long as the original client is still alive
- Client:
  - The client DLLs, PDDirect.dll and PDStandard.dll, are loaded into the process space of your client module
  - PDDirect.dll and PDStandard.dll combine to expose a set of objects and an object model used to perform Palm system and database operations
  - The client DLLs establish and maintain a connection with PDServer using COM/DCOM

Notification Interface

- How does PDServer activate a client module?
- IPDClientNotify specifies a notification interface to be implemented in your conduit
- IPDClientNotify extends part of the conduit API to the client module
- PDServer notifies your conduit when:
  - HotSync is ready for it to run
  - The user selects it from the Custom Menu
- You are not required to use this interface
  - Only ActiveX client modules implement

Client Module Types

- Standard EXE Client
  - Does not implement IPDClientNotify interface
  - PDDirect will activate Standard EXE clients and wait for them to terminate
  - These clients are not required to access PDDirect objects themselves
  - They may spawn other processes/threads which access PDDirect
  - Example use: MS Word, VB, Delphi, IE

Client Module Types

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- ActiveX Client (EXE or DLL)
  - Does implement IPDClientNotify interface
  - PDServer notifies your conduit when:
    - HotSync is ready for it to run
    - The user selects it from the Custom Menu
  - PDDirect uses standard COM mechanisms to get a pointer to your client object that implements the IPDClientNotify interface (the PROGID)
  - Therefore you must build a public object that exposes the interface
  - Example use: Notify enterprise data servers, non-stop processes, etc.

How Do You Use It?

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- A Palm OS® Application
  - PRC with same creator ID as conduit registry entry
- Two keys in the Windows registry
  - PDDirect: String value
    - Standard EXE client will be path to EXE to launch
    - ActiveX client will be notification objects PROGID
  - PDDirectEnable: DWORD value
    - Enable/Disable your conduit next HotSync
- Client Notification Interface
  - Figure out which notification method to use

How Do I Use It?

VB Conduits in Three Easy Steps

1. Easiest to start with a standard EXE, and write your conduit logic
2. Use PDDirect to debug your conduit with your Palm device on-line
3. Convert your conduit to an ActiveX EXE or ActiveX DLL (or leave it as a Standard EXE)
   - Changes the format of your conduit to one you desire
   - Unnecessary if you want a Standard EXE
   - Otherwise, create an object that exposes the IPDClientNotify interface and register the object with COM

Debugging Your Conduit



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- Configure PDServer to launch your IDE
- Since your IDE is a Standard EXE, PDServer will continue to run until you exit your IDE
- Program and debug your application with your Palm device on-line
- Real-time access to Palm data

- You'll see this in the demo

Simple To Use

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```
Private Sub AddToMemo()
    '   Objects
    Dim pDbQuery As New PDDatabaseQuery
    Dim pMemo As PDRecordAdapter
    Dim pUtility As New PDUtility

    '   Variables
    Dim vData As Variant
    Dim nUniqueId As Long

    '   Open the Memo database
    Set pMemo = pDbQuery.OpenDatabase("MemoDB", "PDDirect.PDRecordAdapter")

    '   Create the Memo data
    ReDim vData(0 To 38) As Byte
    pUtility.BSTRToByteArray vData, 0, "Add a new string to the Memo_
        database."

    '   Write a new record
    nUniqueId = 0
    pMemo.Write nUniqueId, 1, eDirty, vData
End Sub
```

DBInfoByName Method



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One read from the Palm...

```
Dim DbQuery as New PDDatabaseQuery
Dim DbInfo as PDDatabaseInfo
Set DbInfo = DbQuery.ReadDbInfoByName("MemoDB")
```

...Returns the DB properties below

- AppInfo block size
- Last backup date
- Memory card number
- Creation date
- Creator Id
- Total data bytes
- Flags
- Index
- Name
- Type
- Excluded from HotSync
- Ram or Rom storage
- Maximum data record size
- Modification count
- Last modification date
- Number of records
- SortInfo block size
- Total size
- Version

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